

THE UNIVERSITY *of* EDINBURGH
School of Engineering

Digital Manufacture 5
Assignment

The feasibility of using digital manufacturing to replace existing brackets for a 1997 Lola T97/20 Indy Lights car with a longer-lasting version

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Contents

1 Reverse Engineering **1**
- 1.1 Implementation of optical scanning process . . . 1
- 1.2 Relative costs . . . 2
- 1.3 Critical Analysis . . . 3

2 Generative Design **4**
- 2.1 Understanding of basic modelling and design strategy . . . 4
- 2.2 Understanding/overview of basic optimisation strategy . . . 4
- 2.3 Overview of optimisation process . . . 4
- 2.4 Reflection of optimisation process . . . 5
- 2.5 Examples of GD . . . 5
- 2.6 Comments on difficulties and outcome . . . 5
- 2.7 Critical analysis . . . 6

3 Additive Manufacturing **7**
- 3.1 FFF build up . . . 7
- 3.2 LPBF build set up . . . 8
- 3.3 Critical analysis . . . 9

4 Communicating Design **10**
- 4.1 Identification of function-critical geometrical features . . . 10
- 4.2 Measurement strategy linked to function . . . 10
- 4.3 Interpretation of dimensional and surface measurement results . . . 11
- 4.4 Application of measurement uncertainty . . . 11
- 4.5 Tolerance decisions informed by simulation . . . 11
- 4.6 Verification feasibility and limitations . . . 12

References **13**

List of Figures

1 RE scanning equipment used in demonstrations . . . 1
2 Generic RE process . . . 1
3 .stl mesh file produced by optical RE process . . . 3
4 Engineering drawing of aluminium bracket and model in Autodesk Fusion . . . 4
5 Finite Element Analysis (FEA) of standard, unmodified aluminium bracket (with exaggerated deformation) . . . 5
6 Optimised aluminium bracket . . . 5
7 Generative design of the aluminium bracket . . . 6
8 Czinger 21C generatively designed brakes . . . 6
9 FFF print orientation options . . . 7
10 FFF print build up of NACA 4418 airfoil turbine blade . . . 8
11 LPBF print orientation in Netfabb . . . 9
12 Exaggerated deflection of bracket in Autodesk Fusion static study . . . 10
13 Standard bracket engineering drawing with tolerances applied . . . 12

List of Tables

1 KPIs of budget and professional RE equipment . . . 1
2 Time and cost breakdown for each stage of RE . . . 2
3 FFF standard bracket orientation comparison . . . 7
4 Comparison table of build statistics for LPBF build of standards and generative brackets . . . 8

AI Statement

I acknowledge the use of AI in the production of this report for the clarification of concepts (ChatGPT), and grammar and clarity improvement based on my independently produced work (Grammarly, Overleaf).

1 Reverse Engineering

1.1 Implementation of optical scanning process

Two optical reverse engineering (RE) products were demonstrated in the lab: the trolley-mounted Faro Quantum^{E} FaroArm (Figure 1a) and the handheld Einstar 3D scanner (Figure 1b) [1] [2]. Their key performance indicators (KPIs) are summarised in Table 1 [3] [4]. For RE, the most relevant KPI's are resolution (minimum measurable feature size), scanning speed, coverage and accessibility, and cost. The FaroArm offers high accuracy (40 micron resolution) and industrial inspection capability at a high cost (£50,000). The Einstar provides lower resolution (100–300 microns) but significantly lower cost (£700–£1,000), making it more suitable for budget and hobby RE tasks.

Figure 1: RE scanning equipment used in demonstrations



(a) Faro Quantum^{E} FaroArm



(b) Einstar 3D scanner

Figure 2: Generic RE process

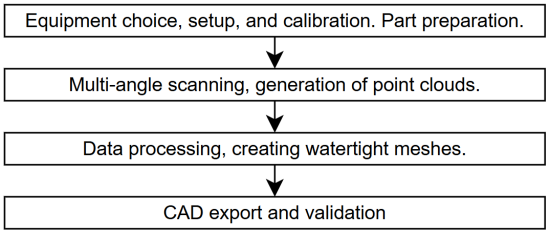


Table 1: KPIs of budget and professional RE equipment

Equipment	Point Resolution	Scanning frequency	Scan Speed	Cost
Faro Quantum^{E} FaroArm	40 microns	300 Hz	600,000 pts/s	£50,000
Einstar 3D scanner	100-300 microns	14 Hz	980,000 pts/s	£700-£1,000

The generic sequence of operations based on Figure 2 is as follows:

1. Setup and Calibration

- Select equipment based on the resolution and accuracy requirements. For the aluminium bracket, the FaroArm would be more appropriate due to the bracket's small size.
- Calibrate the equipment in the software; this is done for both equipment options, the FaroArm calibration is more in-depth as it has higher accuracy and a high number of encoders to calibrate.
- Establish the datum reference plane. The FaroArm probe tip is used to create a plane on a flat surface, so points generated within a user-defined offset, usually ± 1 mm, are ignored.
- Clean and secure the part to prevent movement; glues are more effective than clamps, which would appear as objects in the scan. Apply anti-reflective spray if required (powder causes inaccuracies).

2. Scanning

- Scan visible surfaces, maintaining an optimal working distance, this is 115 mm for the FaroArm and 400 mm for the Einstar. Time-of-flight sensors measure the distance from scanner to object.
- Ensure sufficient coverage of complex features, multiple orientations of the component and separate point clouds may be required. For the Einstar, markers can be placed on similar components to avoid mis-scanning, as it uses previously scanned parts to orient itself, rather than a datum.
- Export point clouds as .txt files, the bracket had 1.9 million data points. These are not meshes.

3. Point Cloud Processing

- Merge multiple scans where necessary, create a watertight mesh using code such as with the TriMesh Python library used in the lab to create a watertight mesh of triangles.
- Meshes can be constructed within the Einstar software, which is more user-friendly. Parts can be manually removed, edited, and analysed.

4. CAD and Validation

- The mesh can be converted into parametric CAD, and features can be reconstructed.
- The scan can be compared to an ideal CAD model to analyse tolerance compliance.
- Simulations such as FEA and natural frequency analysis can then be performed.

1.2 Relative costs

To estimate the relative costs of reverse engineering the aluminium bracket, the following assumptions are made:

- Only a single bracket is being scanned; there is no batch production. Cost per part will scale strongly with the number of parts as efficiency increases as equipment is set up once (economies of scale).
- No internal scanning is required for inspection purposes, such as CT scanning.
- Surface condition is suitable for optical scanning. For example, the aluminium surface is not reflective, which would require preparation for pre-scanning.
- An engineer skilled in RE can be hired with no training required, at the national average of £32 per hour [5].
- Any equipment calibration and software are included in the rental cost.

The cost for the whole operation is equal to the time spent multiplied by the cost of labour, plus the cost of equipment. Based on what was learnt during the lab sessions, the approximate time and cost of each stage is outlined in Table 2, loosely based on the process flowchart (Figure 2).

Table 2: Time and cost breakdown for each stage of RE

Stage	Estimated Time	% Time	Details	Cost	% Cost
Equipment Rental	NA	NA	Estimated weekly rental cost of FaroArm	£1,350	81%
Setup and Fixture	1-2 hrs	14%	Physical setup of equipment, software setup and calibration. Part and scanning surface preparation	£48	2.9%
Scanning	1 hr	10%	Multiple setups and orientations are required, and care should be taken around complex geometry.	£32	1.9%
Data Processing	3 hrs	28.5%	Combining of point clouds, and generating a watertight mesh.	£96	5.7%
CAD Reconstruction	2-4 hrs	28.5%	Any required manual corrections to mesh and features not rectified by data processing.	£96	5.7%
Validation	2 hrs	19%	Deviation analysis using comparison scanned model of another part, or original CAD model.	£48	2.9%

Equipment Costs:
For the reverse engineering a single aluminium bracket, purchasing a FaroArm (£50,000 capital cost) can not be justified economically. The FaroArm is preferred for this application due to a much higher scanning resolution as per Table 2, so rental would be preferred. Several companies offer FaroArm products by quote on request, so defining a definitive cost is difficult. As the total time required ranges from 9 to 12 hours, as per Table 2, it would be sensible to hire the equipment for a week rather than a single day, to allow time for troubleshooting or repeat scans. The total cost is £1,670.

One source in the US rental starts from $500 per day (£370), with discounted rates for weekly rentals [6]. Based on ratios between daily to weekly rentals by other companies, the weekly rental will be between £1,200 and £1,500 [7]. Note that this is above the cost of the Einstar 3D scanner (£700), so if less resolution is required, this scanner should be purchased [8].

Table 2 shows the cost for each part of the process, and relative costs, using the median values for any ranges. It can be seen that the equipment cost dominates, with 81% of the cost. If the Einstar 3D scanner was purchased, the total cost would be reduced by 39%. The data processing and CAD reconstruction accounts for half of the labour cost (11.4%), while scanning is only 1.9% of the total cost.

1.3 Critical Analysis

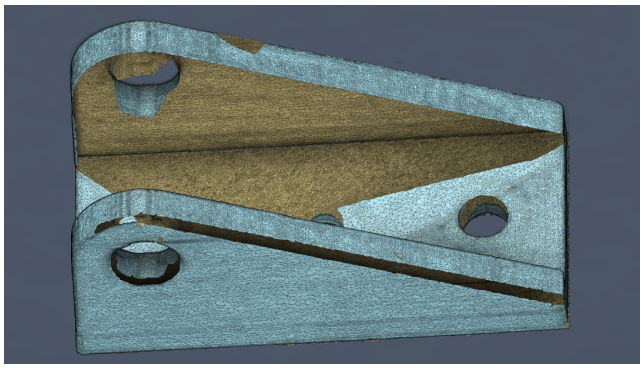
For the aluminium bracket, optical RE systems provide a highly accurate method of recreating the component digitally, where CAD models and drawings are unavailable. The produced point cloud was very dense with 1.9 million data points, providing a detailed surface representation for mesh generation and CAD reconstruction.

Optical RE is particularly useful for deviation analysis, damage inspection and components with complex or organic geometry. However, the bracket to be scanned consists of planar faces, constant-radius curves, and through holes. These features could be measured directly using vernier calipers or a micrometre and remodelled in CAD much quicker and at a far lower cost.

Measuring the thickness of a side panel in multiple places allows an average thickness to be determined. The bracket could likely be measured and modelled within an hour or two with acceptable accuracy, with just a fraction of the cost. If a vernier calliper with a resolution of 10 microns is bought for £170 and the process takes three hours, the total cost would be £266 [9]. At 16% of the optical RE cost, it is hard to justify the optical RE process for this application from a financial point of view.

There are also practical limitations. Optical RE scanning is constrained by line-of-sight, where surfaces, specifically internal, can not be scanned without repositioning of the bracket. This can be seen in Figure 3. Accuracy also depends on several uncertainties, above that of the measuring equipment, such as mis-calibration, user error, and movement of the part during scanning. Reflective surfaces may require matte spray, which introduces further uncertainty.

Figure 3: .stl mesh file produced by optical RE process



When converting point clouds to a watertight mesh, the number of faces can be reduced to decrease computational power and memory usage during simulations for example, at the expense of detail. This negates some of the benefits of a very high scan resolution, and the need for a FaroArm for example, if the results from the Einstar 3D scanner are going to be similar once the .stl model file is produced.

Further work would include comparing the optical RE produced CAD model to a CAD model using manual measurements collected using a vernier caliper or micrometer as previously described. This would include repeat scans and deviation analysis between the models, to determine whether optical RE gives a representative improvement for this type of geometry, to determine the uncertainties involved, and generate an accurate set of dimensions to be used further in the project.

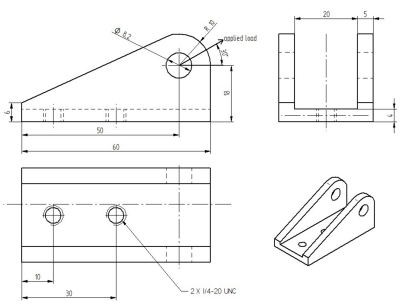
To conclude, while optical RE is a technically impressive and capable technology, it is not recommended for reverse engineering the aluminium bracket, as manual metrology is significantly more cost-effective while being sufficiently accurate. Optical RE is better justified for organic, complex, or damaged components, or for inspection tasks where full surface comparison is required.

2 Generative Design

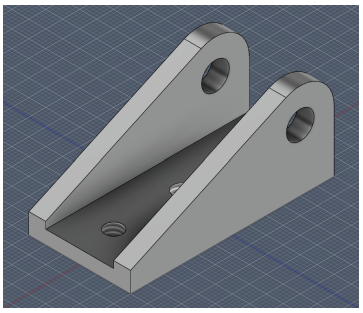
2.1 Understanding of basic modelling and design strategy

Using Autodesk Fusion, the aluminium bracket was modelled according to the drawing shown by Figure 4a [10]. The threads were modelled as the ANSI Unified Screw Threads, 1/4-20 UNC. The material was set to Aluminium 6061, a common lightweight but high-strength aluminium alloy, giving a part mass of 41.106 grams with a density of 0.0027g/mm^3 [11].

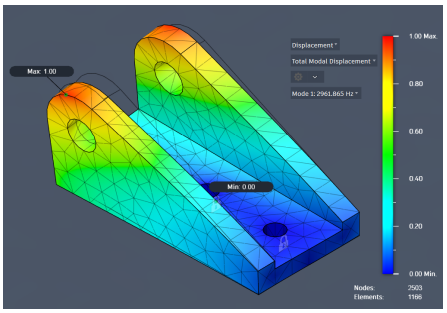
Figure 4: Engineering drawing of aluminium bracket and model in Autodesk Fusion



(a) Engineering drawing of aluminium bracket



(b) Aluminium bracket modelled in Autodesk Fusion



(c) First modal frequency simulation

The first and second natural frequencies were obtained by setting up a model frequencies simulation for the first two modes, with all other settings left as default. A fixed constraint was applied to the two threaded holes, and a 3 kN load applied to the side panel holes at an angle of 30 degrees to the horizontal plane. The first and second natural frequencies were 2961.865 Hz (Figure 4c) and 3845.219 Hz. These frequencies should be compared to those produced by the car to avoid resonant frequencies.

2.2 Understanding/overview of basic optimisation strategy

Optimisation is where the ”best” solution to a problem is found using a computational design methodology, usually to maximise or minimise a parameter under defined objectives and constraints. In an engineering context, this can be minimising mass for a specified strength requirement, or to maximise stiffness.

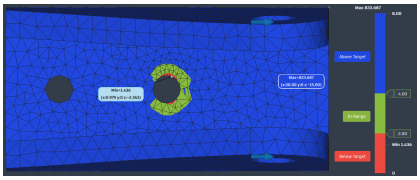
Unconstrained optimisation is when the searching algorithm is free to minimise or maximise the objectives with no restrictions, which can lead to problems. For example, to minimise mass with no stress or manufacturability constraints, the optimal solution would be to remove all material, which would be physically meaningless, and therefore constraints are almost always used in engineering design.

Generative design is almost always a constrained optimisation problem, satisfying structural performance criteria including safety factors, manufacturing constraints and geometrical constraints. In the case of the aluminium bracket, there must be adequte flat surface area for the bolts to be seated and tightened with tools, and no geometry interfering with any bolt through the threaded or unthreaded holes.

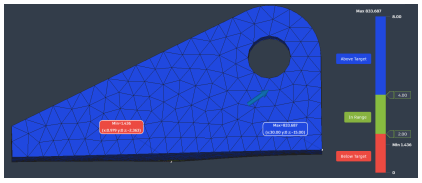
2.3 Overview of optimisation process

A static stress simulation study within Fusion was used to conduct an FEA analysis of the unmodified bracket, where the 3 kN load is applied to the side panel holes at a 30-degree angle to the horizontal plane. It was found that the minimum safety factor was 1.436 around the central threaded hole on the bottom surface, below the target of 4, as shown in Figure 5. Below target stresses are on the top and bottom surfaces, rather than in the middle on the bottom plate, suggesting that a thicker base plate may be necessary. A finer mesh than the default was used, which exposed this area of weakness, which will be the focus of optimisation. The maximum safety factor was 833.687 at the far corner in Figure 4b, suggesting that material may be removed here, or side panels made thinner, with no functional performance cost.

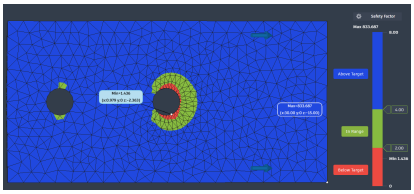
Figure 5: Finite Element Analysis (FEA) of standard, unmodified aluminium bracket (with exaggerated deformation)



(a) FEA top view



(b) FEA side view

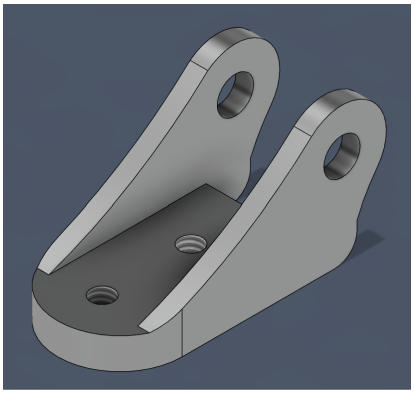


(c) FEA bottom view

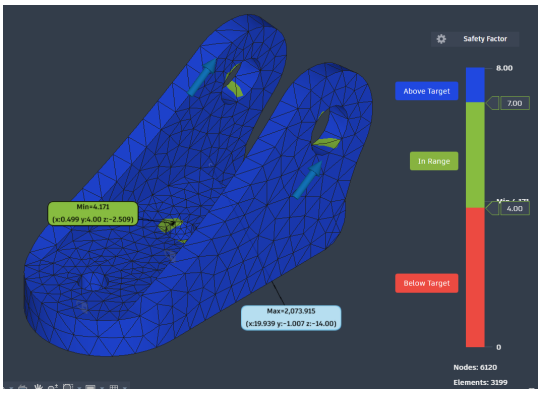
2.4 Reflection of optimisation process

This optimisation process was done manually, rather than by an automatic computational process. The first stage was to increase the thickness of the base plate until the safety factor of 4 was achieved. An increase from 4 mm to 7 mm gave a safety factor of 4.049 (Figure 6a) with mass 55.909 grams. Increasing the thickness of the side panels decreased the safety factor around the holes. 1 mm thinner side panels increased the safety factor to 4.299, reducing mass to 49.108 grams (Figure 6b). Removing material on sharp corners and areas identified as low stress, the safety factor is 4.171, the mass is lowered to 34.938 grams, 15.0% lower than the original. From this stage of optimisation, however, it became difficult to intuitively remove material while maintaining a safety factor of 4 after the obvious modifications.

Figure 6: Optimised aluminium bracket



(a) 3D CAD model



(b) FEA static study

2.5 Examples of GD

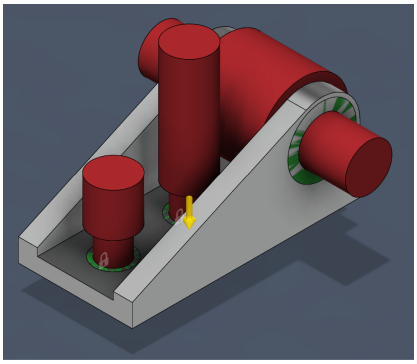
Generative designs were generated using Autodesk Fusion with a minimising mass target. As shown in Figure 7a, the areas around the bolt holds were set as preserved geometry (green) to allow a flat surface for bolts to be seated, and obstacle geometry (red) was set up to define areas where geometry could not be generated. These were spaces where bolts would be located, would need to pass through during installation, or would be space for tool access. The loading conditions and constraints as per 2.3 were applied. A generative design was produced starting from the standard aluminium bracket (Figure 4b), and the optimised aluminium bracket (Figure 6a), as shown in Figures 7b and 7c after 25 iterations.

The first generative design had a mass of 47 grams based on the original bracket, an increase of 12.5%. The second generative based on the manually optimised bracket had a mass of just 28 grams, a decrease of 31.9% over the original, and 19.9% compared to the manually optimised bracket.

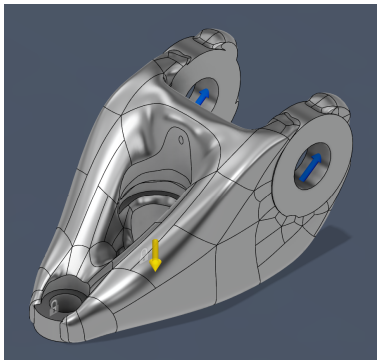
2.6 Comments on difficulties and outcome

During the manual optimisation process, it was a tedious task to find the optimum combination of base and side panel thicknesses, with just two parameters to adjust. Subsequently, finding where material could be removed took a few hours, with nearly twenty FEA simulations completed. Generative design can save time on this manual work, especially for more complex components where perfection would take a near infinite length of time. However, starting from the original bracket, the mass increased

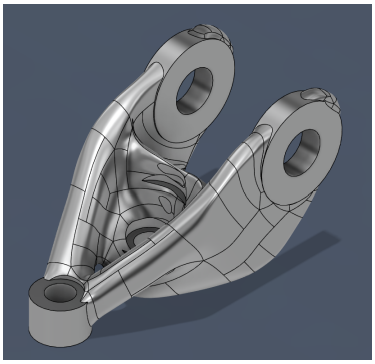
Figure 7: Generative design of the aluminium bracket



(a) Setup with preserve and obstacle geometry



(b) Generative design from original aluminium bracket



(c) Generative design from original aluminium bracket

to obtain the required safety factor. Mass decreased substantially when starting from the optimised bracket, suggesting that manual optimisation using intuition followed by generative design may work best. However, when FEA was performed on the generated designs to validate their strength, it was found that they only had a safety factor of 1.044 and 2.109, making them unusable.

The parts were generated for additive manufacturing and milling, with the unrestricted setting on. If additive manufacturing is used, the holes should be drilled afterwards. With milling, there would be more material compared to additive manufacture as the starting material would be a block, rather than sheet metal, which could be used for the original design.

2.7 Critical analysis

Generative design provides a powerful method for optimising structural components, particularly where intuition alone becomes inefficient. As demonstrated in the manual optimisation process, identifying the optimal combination of parameters required numerous time-consuming FEA simulations. Generative design automates this process by performing stress analysis at each iteration; however, the results depend strongly on the accuracy of the underlying simulation model. The generative designs produced in section 2.5 achieved safety factors of only 1.044 and 2.109 when validated through a separate FEA analysis, so the desired safety factor must be verified outside the generative design workspace.

The generative design also assumes ideal loading conditions. The generated bracket was designed for a 3 kN force applied at 30 degrees, whereas the real application will involve cyclic and potentially variable loading conditions. If the loading direction or magnitude differs, the optimised structure may not perform as intended. In practice, this is addressed by applying a safety factor and validating the design through additional simulations and mechanical testing.

Generative design is particularly effective for identifying efficient load paths, which can inform improved conventional designs. Although the generated geometries are often complex and require additive manufacturing, the generated models can also be used to inform components manufactured using conventional methods.

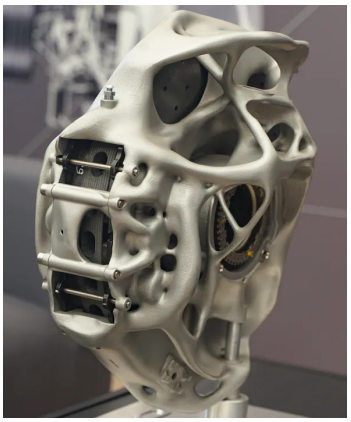


Figure 8: Czinger 21C generatively designed brakes

The technology is already used in high-performance industries such as automotive and aerospace. For example, the Czinger 21C hypercar uses generatively designed suspension and braking components manufactured using additive processes (Figure 8) [12] [13]. However, these applications are typically low-volume and high-cost, so generative design is not yet widely used in mass-produced vehicles, as they are not high-performance applications.

3 Additive Manufacturing

3.1 FFF build up

The fused filament fabrication (FFF) build was prepared using an .stl file exported from the Autodesk Fusion model (Figure 4b) and imported into PrusaSlicer [14]. The bracket is to be printed on the Prusa MINI & MINI+ Input Shaper printer, with Prusament PLA polymer filament [15].

Several geometric features present challenges for FFF printing of the bracket. Most critical are the holes and overhanging surfaces, which are problematic due to the likelihood of sagging and creep while the printed filament is cooling. With holes in both the side panels and base plate, one pair will always be compromised in terms of geometrical accuracy, regardless of orientation. This may be avoided by making the hole a teardrop shape [16]. Two possible orientations were considered (Figure 9):

Upright Orientation (Figure 9a): In this orientation, the base plate is placed on the print bed, providing:

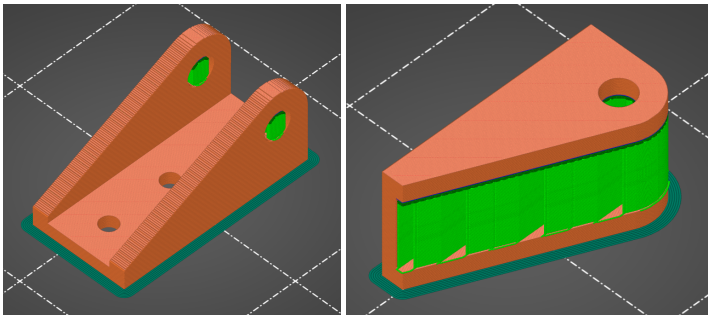
- A flat and smooth base plate surface, ideal as it is the mounting surface in contact with the car.
- Low support requirement: only required for side panel holes, reducing filament volume required. May have poorer surface finish and dimensional accuracy after removal.
- Dimensional symmetry is achieved. Slope of side panels will have a rough surface, but can be mitigated with ironing [17].

Side Orientation (Figure 9b): Printing the bracket on its side improves the circular accuracy of the side-panel holes as they are aligned with the build direction. However, there are a few disadvantages:

- Large overhanging regions requiring significant support material.
- Increased print time and material usage.
- Potential distortion of the suspended side panel due to cooling shrinkage.

Build differences are shown in Table 3, where the side orientation requires over 20 times more support material and a longer print time. Strength is not optimised here, as the print is for a visual mock up.

Figure 9: FFF print orientation options



(a) Upright orientation (b) Side orientation

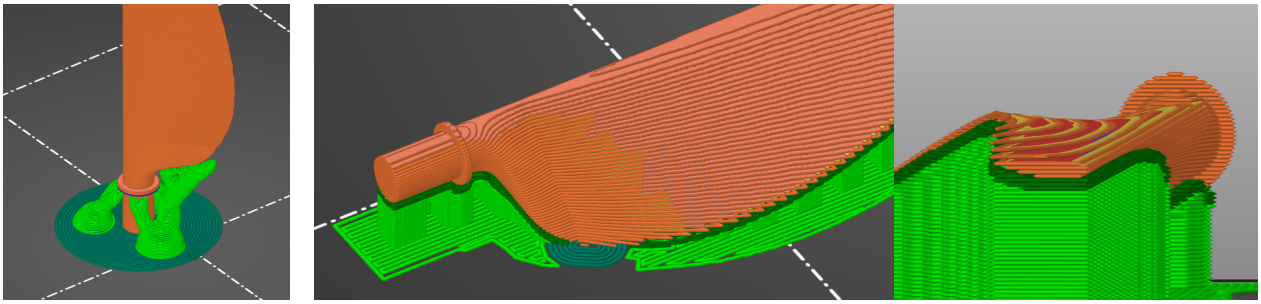
Table 3: FFF standard bracket orientation comparison

Orientation	Upright (9a)	Side (9b)
Filament	11.81 g	15.85 g
Print Time	1 hr 24 min	1 hr 48 min
Support Material	0.15 g	3.50 g
Ironing Time	47 min	1 hr 2 min

For the second model, a NACA 4418 turbine blade was selected due to its complex aerodynamic shape [18]. Challenges include the thin trailing edge, root mounting hole and airfoil curvature, where the staircase effect can be problematic for surface accuracy and aerodynamic performance. Vertical and flat orientations are considered again. In the vertical orientation, the build height is large (157.5 mm), but the gradual change in airfoil profile means successive layers follow the surface closely, minimising the staircase effect. Wide supports are required at the base to prevent the blade from moving during printing (Figure 10a).

Printing the blade flat marginally reduces build time (43 to 39 minutes), and the number of layers significantly (788 to just 53), but introduces significant stepping across the aerodynamic surface (Figure 10b and 10c), resulting in very poor surface finish and geometrical accuracy. The volume of support material also increases, from 0.17 to 2.48 grams, then accounting for 40% of the total material used. Therefore, the vertical orientation is preferred, despite the greater build height, as it maintains the aerodynamic shape and geometrical accuracy of the blade and reduces material usage.

Figure 10: FFF print build up of NACA 4418 airfoil turbine blade



(a) Vertical orientation supports (b) Flat orientation root (c) Flat orientation blade tip

3.2 LPBF build set up

The laser powder bed fusion (LPBF) build was prepared in Autodesk Netfabb using the .stl model of the standard and generative brackets bracket (Figure 7c) [19]. Both are to be printed on the Aconity Mini L-PBF printer [20]. The aim of the build setup is to maximise structural performance while maintaining sufficient dimensional accuracy for assembly.

Design Modifications for LPBF: Metal LPBF parts typically require post-processing, including heat treatment for residual stress removal, and machining due to high surface roughnesses. In anticipation, an additional 0.5 mm of material is added to the dimensionally important and functional surfaces, that being the inner faces of the side panels and the top and bottom of the base plate. On the generative design these are added to the flat mounting regions. LPBF technology does not currently have adequetly fine accuracy for printing internal threads, therefore all holes were removed from both models, as they can be easily machined after printing [21].

Build Orientation: Build orientation strongly influences mechanical properties in LPBF due to the anisotropic properties of the layer-by-layer build-up. Components are stronger in the XY plane along the layer lengths, while they are weaker in the Z direction, due to factors such as imperfect fusion and voids. To improve load-bearing performance and fatigue life, both brackets were oriented 30 degrees from the horizontal (Figure 11a and 11b), to align with the load direction shown in Figure 4a.

This orientation also reduces the effective cross-sectional area of each layer, helping to minimise residual thermal stresses and distortion during cooling. However, it requires additional support and produces a rough surface on the underside of the base plate for the standard bracket. This is acceptable because machining allowances were included. To minimise warping, the brackets are kept upright to keep the base plate near the print bed, minimise distortion and residual stresses, decrease the mass of supports required, and decrease recoater interference.

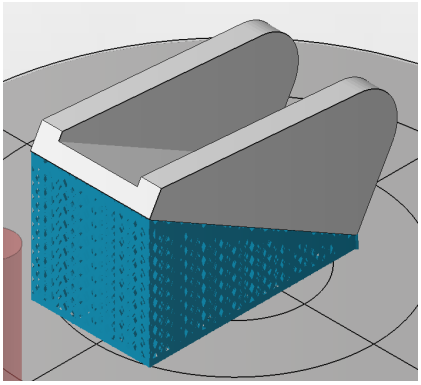
Support Strategy: Support structures are required to prevent thermal distortion and overhangs deformation, which can not be reliably supported by the power bed alone. They needed to be stiff and continuous, but not too difficult to remove, thus the m-pbf_light support script is chosen for both brackets, seen in Figure 11a and 11b, with additional manual supports added beneath flat surfaces on the generative design.

The laser path at approximately half height is shown by Figure 11c. Table 4 shows the build statistics of each part build. The generative bracket uses less material, has a lower build height and shorter build time, which shows that the optimised geometry is more efficient for manufacturing.

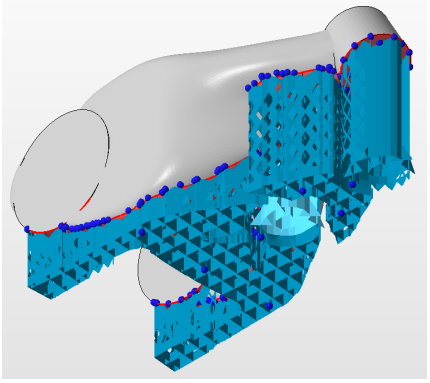
Table 4: Comparison table of build statistics for LPBF build of standards and generative brackets

Part	Volume	Support	Build Height	Build Time	Layer Count
Standard Bracket	16.36 cm^3	2.81 cm^3	37.73 mm	5 hr 42 min	1259
Generative Bracket	11.27 cm^3	793.73 mm^3	26.49	3 hr 56 min	884

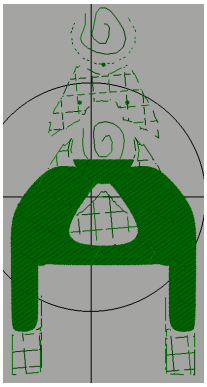
Figure 11: LPBF print orientation in Netfabb



(a) Standard bracket with supports



(b) Generative bracket with supports



(c) Generative bracket laser path

3.3 Critical analysis

For the application of a limited production run of metal brackets for use on the Lola T97/20, LPBF additive manufacture (AM) would be very valuable for manufacturing the generative design bracket (Figure 7c). LPBF has a strong advantage in that it can produce organic shapes and curves, such as those on the generative design. For the component shown in Figure 7c, it is however, likely possible to produce the part using 5-axis CNC milling. For the standard bracket, due to its basic design, it is possible to construct it using sheet metal for the base and two side panels welded together with holes machined in later.

Comparing the options of CNC milling and AM for the generative design bracket, CNC milling is more expensive than additive manufacturing for small batches due to tooling costs, but it is more scalable [22]. It would be beneficial to conduct a cost analysis of the two methods. AM is typically cheaper as material waste and excess machining hours are avoided. Due to the poor tolerating capabilities of AM, important surfaces will require surface finishing as discussed in 3.2, which is likely to be done via milling anyway, albeit a 3-axis CNC machine would suffice, so this is a significant drawback. If there were complex internal geometry, then AM would be one of the only viable options. AM is highly advantageous for rapid prototyping with short lead times and beneficial for testing design alterations. This can be done with the FFF PLA brackets, although the mechanical properties will not be representative of the metal bracket.

Despite its advantages, LPBF has some important limitations. Mechanical properties are anisotropic, so strength and fatigue performance are orientation dependent, despite the orientation choices in 3.2. Residual thermal stresses, distortion, surface roughness, and the need for support removal all reduce dimensional accuracy, increasing post-processing and why extra material for machining was added. Still requiring traditional machining is a significant drawback. However, AM is used successfully for high-performance parts, such as in the Czinger 21C hypercar, 747 class engine compressor spool, and the fuel impeller used in the space shuttle main engine [12] [23].

Overall, while AM has its limitations, including anisotropy and the requirement for post-processing, the advantages of the ability to produce mass-saving complex shapes that convential manufacturing can not make LPBF suitable for manufacturing the generative bracket. Mechanical testing should be conducted before use to verify its mechanical properties, due to the number of defects that can occur, primarily relating to the microstructure and porosity. For the standard bracket, conventional manufacture rather than AM would likely be cheaper, simpler and more dimensionally reliable. FFF is suitable for rapid prototyping, as is LPBF, although the manufacturing time is much longer.

4 Communicating Design

4.1 Identification of function-critical geometrical features

Based on the FEA results discussed in section 2.3 and shown in Figure 12, the side panels are pulled inwards by the 3 kN tension force by a maximum of 0.067 mm around the side panel holes. The FEA results showed that the highest stress occurred at the hole located centrally on the base plate. The locations and sizes of the holes, as well as the thickness of the side panels, which distribute the load and transfer the load to that central hole, are of high importance. As the side panels deform inwards under load, the holes also bend inwards, so the choice of tolerance fit for the holes is important, especially if plastic deformation takes place, for functionality and ease of fit of bolts. The internal space between side panels is also important, so that any fixtures can fit within.

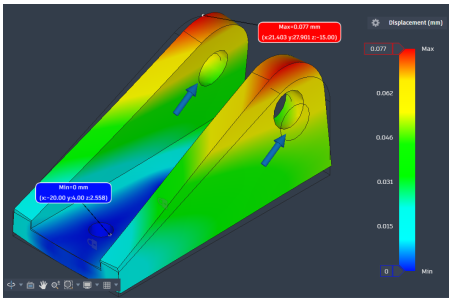


Figure 12: Exaggerated deflection of bracket in Autodesk Fusion static study

The length and height dimensions of the bracket panels are of low importance, as the FEA results showed very high safety factor values here in the extremities (up to 833), as there is no load transfer here.

4.2 Measurement strategy linked to function

The equipment demonstrated in the lab was the Faro Quantum^{E} FaroArm fitted with a contact probe tip, and the Faro Vantage Laser Tracker. The FaroArm is a contact measurement machine (CMM) and works by placing the probe against a surface and pulling the trigger to register a point in space. Repeating at another point generates a line, and once again gives a plane. The contact probe compensates for the radius, and the software can be told which side the plane or hole is on. The FaroArm is not a laser scanner like the one described in the Reverse Engineering section (Section 1), and only collects data points when the trigger is pulled. The probe is more accurate than the Vantage laser scanner with $\pm$46 micron worse case accuracy, compared to $\pm$78 microns. Before scanning, the datum can be set up by creating a plane with three points on the worktop. If the surface on which the measured object is not flat, this will induce error in the measurements.

The purpose of the FaroArm is to verify tolerances on manufactured components, so it is suited to the purpose of measuring tolerances on the aluminium bracket, and the demonstration piece measured in the lab. The 'length of features' tool is used to measure the thickness between plates, the 'circle' tool is useful for measuring bores and can be used to measure cylindrical objects as well, such as the handle on the demonstration plate. As points are measured by contact and not by optical lasers, capturing complex and organic shapes, such as those printed using LPBF, can be very difficult, or time-consuming at the least. Internal geometry can be difficult or impossible to measure.

The measurement process goes as follows: starting from an engineering drawing, the tolerances of interest are identified, and these dimensions are then measured on the physical piece, data is built up within the machine's software. A CAD file of the nominal component can be imported into the software to perform data analysis. The distance between the CAD component and the measured data points is signified by colour, to highlight areas where the physical part is out of tolerance or has been damaged.

A similar process occurs with the Faro Vantage Laser Tracker, although this is suited to larger objects, as it has a 80 metre range spherically. It uses a 38.1 diameter locating sphere do fine details are diffilcult to capture. Also, as the laser tracker itself is fixed in position (if it were moved, errors would be induced), it is not possible to measure around corners without the handheld 6 DoF SMR probe, which will induce additional uncertainties. This piece of equipment is very expensive at £100,000 for the 2021 model and £10,000 to purchase the software outright.

4.3 Interpretation of dimensional and surface measurement results

Although no physical measurement results were taken in this assignment, probable dimensional or surface measurement results can still be interpreted in the context of the bracket's function. For example, if probing results showed that the side-panel holes were displaced inwards relative to nominal, or that the spacing between the side panels was smaller than intended, assembly of any mating part and insertion of bolts would become more difficult due to the angle, effectively making the holes smaller. This is important because the FEA already showed inward elastic movement of the side panels under the 3 kN load, so any manufactured deviation in the same direction would add further deformation. Additionally, a deviation around the central threaded hole on the base plate is more significant than a similar error at the free ends of the bracket, because this region was identified as where the principal stress concentration occurs.

Surface measurement results such as roughness, waviness, or support-removal marks on the base mounting face, reduce true contact area and can increase local contact pressure. Of these, only waviness can be picked up by the FaroArm probe due to its 3 mm diameter. Poor surface condition around the holes may cause poor bolt seating, fretting, and increase the likelihood of local crack initiation. Small deviations at the extremities of the bracket are less serious as these regions are lightly loaded.

4.4 Application of measurement uncertainty

Measurement uncertainty should not be treated only as a result of machine accuracy, but as the combined effect of all factors influencing the measurement process. For the bracket, these include probe accuracy, calibration state, datum establishment on the base face, fixture stability, accessibility of the probe in and around threaded holes, probe placement from the operator, burrs or roughness on the measured surfaces, and temperature mismatches between the part and the environment. This is especially relevant here because the FaroArm probe is intended for discrete dimensional verification of planes, circles and distances, with a worst case accuracy of ± 46 microns. Therefore, when a circle is fitted to a hole or a plane is fitted to a rough LPBF surface for example, the reported value is only an estimate of the true geometry rather than an exact value; a constant radius is assumed on holes.

Practically, uncertainty should be used to support acceptance decisions. A bracket should only be accepted when the measured deviation, plus its uncertainty, still lies within the chosen tolerance band; similarly, it should only be rejected when the measured deviation, minus its uncertainty, lies outside the tolerance band. Measurements falling between these two cases should be treated as inconclusive and would require repeat measurement or a revised datum setup. For the aluminium bracket, that matters most for the side-panel spacing, side-hole position and base thickness, because these directly affect fit, load transfer and safety factor. A measured error that appears small in isolation may still be unacceptable once uncertainty is considered, particularly since the simulated inward deflection of 0.067 mm is of the same order as plausible manufacturing and measurement variation. The uncertainty, therefore, needs to be accounted for when deciding how tight or loose the tolerance margin can be while still preserving reliable assembly and structural performance.

4.5 Tolerance decisions informed by simulation

The simulation results indicate that the tightest tolerances should be assigned to the dimensions that most strongly influence stiffness, hole alignment and load transfer. The minimum safety factor of 1.436 occurred around the central threaded hole on the base plate (Section 2.3), and increasing the base thickness to 7 mm increased the safety factor to 4.049. This shows that the base region is highly sensitive to geometry and should be controlled more tightly than the outer profile or free edges. A suitable datum would therefore be the bottom mounting face as datum A, with the inner surface of the left side panel and the back of the bracket, i.e the surface at the end of side panels on the end of the holes, used as secondary and tertiary datums B and C. Relative to this datum system, the side-panel holes can be controlled using a positional tolerance from datum A and/or C, while the internal spacing between the side panels can be controlled using a size tolerance from datum B and a parallelism requirement.

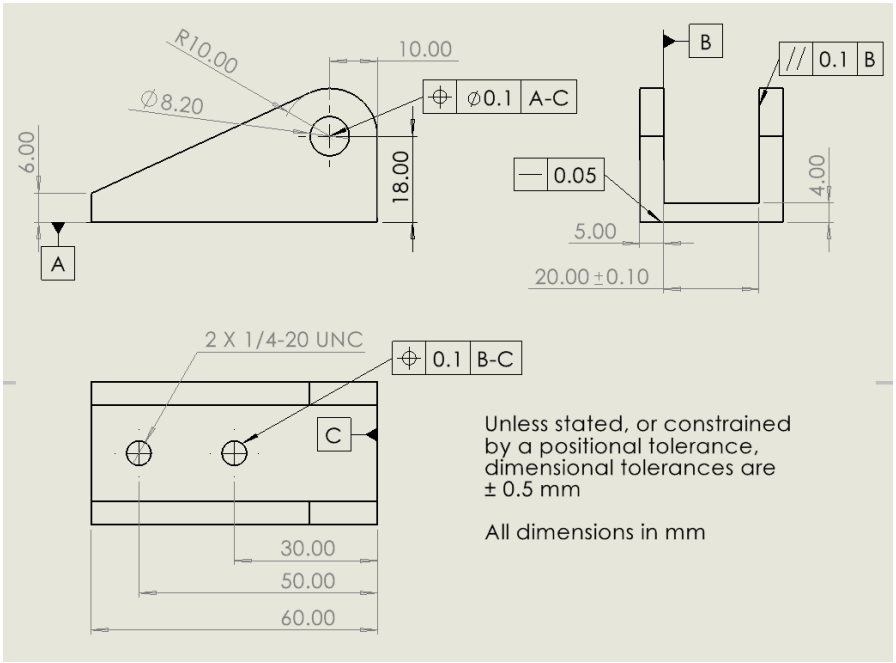
A suitable tolerance specification would be a positional tolerance of ∅ 0.20 mm for each side-panel hole relative to datums A, B and C, so that the hole axes remain correctly located with respect to the mounting face, the inner locating face of the bracket, and the rear locating face. This means that the true position of the hole centre must lie within a cylindrical zone of diameter 0.20 mm around that ideal position.

For the internal dimensions, the spacing between the side panels can be controlled using a bilateral size tolerance of ± 0.10 mm, with the opposing inner face from datum B assigned a parallelism tolerance of 0.10 mm so that the gap remains uniform from the top to the bottom of the side panels. The bottom mounting face should also be assigned a flatness tolerance of 0.05 mm, as this is the primary datum and must maintain good contact with the car and avoid local rocking under load.

These values are justified because they keep manufacturing variation small relative to the simulated deformation at the side holes, while recognising that the highest stresses are transmitted through the base and central hole region. The outer non-functional dimensions of the bracket can be given a looser profile tolerance, for example 0.50 mm, because the FEA showed that these areas have very high safety factors and not on the main load path. In this way, the simulation informs which tolerances are critical to function and which can be relaxed without any compromise to performance.

The dimensions and applied tolerances are shown in Figure 13, produced using the original aluminium bracket modelled in SolidWorks, with tolerancing in accordance with ISO 1101-2017 [24]. Relative to the pin or bolt being inserted into the side panel holes, which is unknown, the hole should be a clearance fit such that the pin or bolt is insertable even if the deformation found in the simulation is present.

Figure 13: Standard bracket engineering drawing with tolerances applied



4.6 Verification feasibility and limitations

As the probe tip on the FaroArm has a 3 mm radius, this adds uncertainty to holes smaller than this being measured. Additonally it cannot measure surface roughness as the curve of the probe tip is large relative to the scale of surface roughness, due to the uncertainty of ± 46 microns, but could pick up a wavy form.

The probe tip measurement technique is best suited for measuring planes, holes and distances, rather than organic shapes. Therefore, objects with organic shapes can not be measured accurately without a near infinite number of data points recorded. Here, the laser scanning technique used in the Reverse Engineering section would be applicable.

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